Modeling Human Activity States Using Hidden Markov Models

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GitHub repository: https://github.com/f-Ayuk/favor_hmm.git

1. Background and motivation

Our group's use case is a lightweight smartphone activity monitor that can label everyday movement as still, standing, walking, or jumping. Recognizing these states can support personal activity logging and context-aware mobile services without requiring specialized wearable equipment. A Hidden Markov Model (HMM) is appropriate because motion is measured as a sequence: the current activity is not observed directly, and nearby time windows are likely to share the same state. The project therefore tests whether features from phone accelerometer recordings can identify these four controlled activities.

2. Data collection and pre-processing steps

The supplied dataset contains 79 CSV recordings organized by activity: 17 still, 20 standing, 21 walking, and 21 jumping. The dataset summary reports approximately 559 seconds of recording in total. The sampling-rate report shows an original rate of about 125 Hz. No gyroscope channels were detected in these recordings, so the implemented model uses accelerometer data only.

Collection video links

Activity	Collection video
Still	Open still video
Standing	Open standing video
Walking	Open walking video
Jumping	Open jumping video

The links above were supplied for the required five-second collection clips. The report records the links but does not independently verify video duration, framing, face visibility, or lighting.

Pre-processing and feature construction

Time values were parsed, non-finite and non-increasing samples were removed, and each recording was linearly interpolated onto a uniform 50 Hz time grid. Each sequence was divided into two-second windows (100 resampled observations) with a one-second stride, giving 50% overlap. Features were standardized with a StandardScaler fitted on training windows only.

For each of the three accelerometer axes and the acceleration magnitude, the code computes mean, standard deviation, minimum, maximum, root-mean-square, range, skewness, kurtosis, zero-crossing rate, interquartile range, energy, dominant frequency, spectral centroid, and spectral entropy. The spectral features are derived from a Welch power spectral density estimate. Together these form 56 features per window. Although the assignment example mentions gyroscope features, none were present in the supplied recordings.

3. HMM setup and implementation details

The model is a four-state Gaussian HMM. Each state corresponds to one activity, and each observation is a standardized 56-dimensional feature vector. The emission distributions use diagonal covariance, which models per-feature variance without estimating every cross-feature covariance.

Element	Implementation
Hidden states (Z)	Still, standing, walking, and jumping.
Observations (X)	One 56-feature vector per two-second window.
Transition probabilities (A)	Initialized with 0.95 self-transition probability and 0.05/3 for each other state; learned during training.
Emission probabilities (B)	Diagonal-covariance Gaussian distributions over the continuous feature vectors.
Initial probabilities (pi)	Uniform initialization across the four states.

Training and decoding

Training starts from label-based state means and variances, then uses Baum–Welch (expectation-maximization) for up to 200 iterations with a 10^{-4} convergence tolerance. Per-recording sequence lengths are passed to preserve recording boundaries. Viterbi decoding produces the most likely state sequence. This combination suits continuous sensor features and temporal dependence: sensitivity measures missed activity windows, specificity measures correct rejection of other states, and accuracy summarizes correct labels overall.

4. Results and interpretation (including visualizations and metrics)

Evaluation held out the last available recording for each activity and trained on the remaining recordings (76 training recordings, four held-out recordings). The held-out recordings were still part of the supplied dataset; the project does not document a separate participant, collection session, or environment. Thus, this is a recording-level test, not a verified cross-person or cross-setting generalization test.

Held-out classification metrics

State (activity)	Number of samples*	Sensitivity	Specificity	Overall accuracy
Still	5	100%	100%	100%
Standing	6	100%	100%	100%
Walking	6	100%	100%	100%
Jumping	6	100%	100%	100%
Overall	**23**	**100%**	**100%**	**100%**

*Samples are two-second windows, not raw sensor readings. The overall accuracy is 23/23 windows. Because adjacent windows overlap by 50% and come from only four recordings, they are not 23 independent trials.

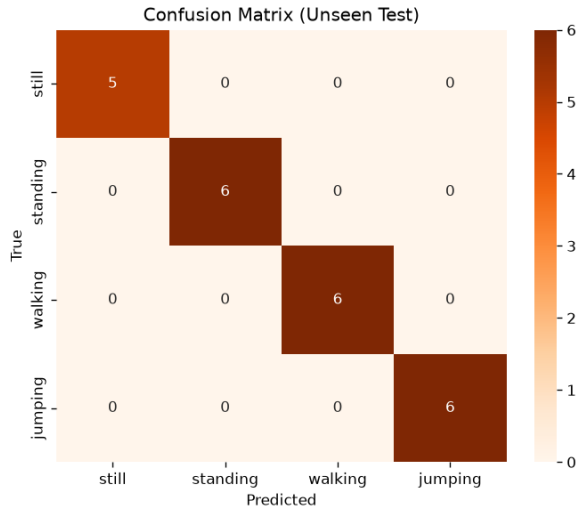


Figure 1. Confusion matrix for the held-out windows. All 23 predictions lie on the diagonal.

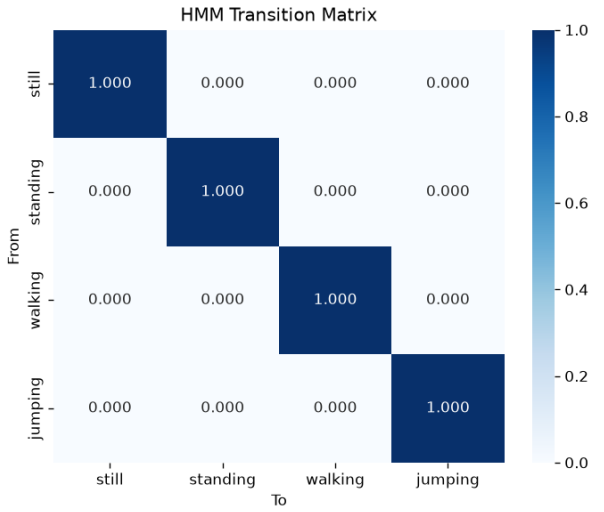


Figure 2. Learned transition matrix. Values round to 1.000 on the diagonal and 0.000 off it.

The test confusion matrix contains no observed errors. The learned transition matrix is almost entirely self-persistence. This is consistent with recordings that each contain one labeled action, but it does not demonstrate realistic transitions between activities.

Results and interpretation — visualizations

The feature plots show examples of how motion intensity varies across the activity labels. Still windows have low spread in these features, while jumping generally has larger variability. These plots are descriptive; they are not a separate test of generalization.

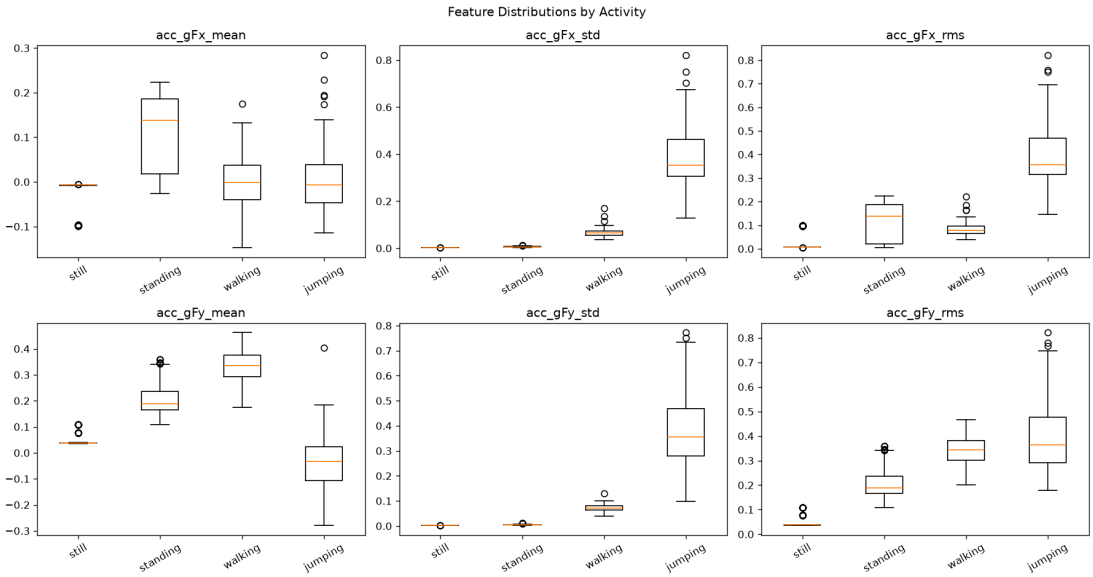


Figure 3. Distributions of selected accelerometer mean, standard deviation, and RMS features by activity.

Each decoded plot below shows the true label and Viterbi prediction for one held-out recording. The traces overlap because every window in each selected recording was assigned its correct single activity.

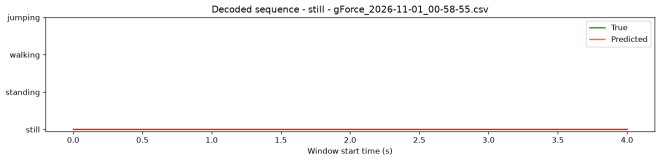


Figure 4a. Still — 5 windows; true and predicted labels overlap.

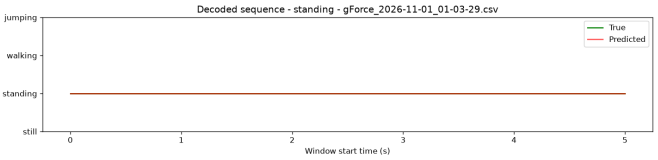


Figure 4b. Standing — 6 windows; true and predicted labels overlap.

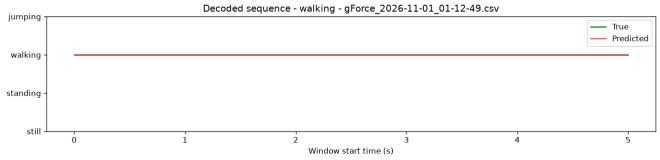


Figure 4c. Walking — 6 windows; true and predicted labels overlap.

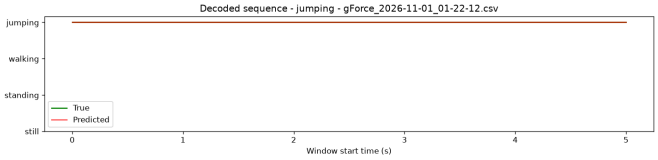


Figure 4d. Jumping — 6 windows; true and predicted labels overlap.

These single-action recordings contain no within-recording class changes. A continuous recording that moves through multiple activities is needed to evaluate transition timing and boundary errors.

5. Discussion and conclusion.

Activity separability and evaluation

The perfect score means no activity was harder on this particular test set: sensitivity and specificity are 100% for all four labels. It does not establish that the classes are equally easy in general. The feature distributions suggest that still and jumping can have distinct motion levels, while standing and walking may be more challenging when posture, phone orientation, or movement intensity varies. More independent test recordings are required to rank class difficulty reliably.

Transitions, noise, and sampling

High self-transition probabilities are reasonable for sustained activities within a short window. However, the near-zero learned cross-state probabilities likely reflect the way the dataset was recorded as separate single-activity files; they should not be interpreted as a realistic everyday activity-transition model. Resampling from about 125 Hz to 50 Hz creates a consistent input rate, but the implementation uses linear interpolation without an explicit anti-alias filter. Phone placement, orientation, sensor noise, and short recordings may all affect the extracted features. Accelerometer-only input also omits rotational motion that a gyroscope could capture.

Limitations and improvements

The test consists of one recording per class and 23 overlapping windows, all from the same supplied collection. Participant and environment independence are not established. The next evaluation should use more participants and sessions, split by participant or session, and include continuous mixed-activity recordings. Further improvements include adding gyroscope data, applying a suitable anti-alias filter before downsampling, evaluating repeated cross-validation splits, and reporting confidence intervals and class-specific error counts.

Conclusion

This project implements a four-state Gaussian HMM using accelerometer window features and Viterbi decoding. It correctly classified all 23 held-out windows in the current evaluation, and the required confusion, transition, feature, and decoded-sequence figures are included. The result is promising for controlled recordings, but broader claims require an independent, transition-rich test set and sensor data that include gyroscope measurements.